



DataCo Global

Phase 3 | Data Understanding & Metadata Documentation



Source Datasets	2 (DataCoSupplyChainDataset.csv, tokenized_access_logs.csv)
Business Fields	61 (53 main + 8 clickstream)
Business Domains	7 (Customer, Product, Order, Shipping, Financial, Geospatial, Clickstream)
Document Status	Metadata Documentation Complete

1. Purpose

This document defines every field in the DataCo Global supply chain datasets. It serves as the single source of truth for column-level metadata, providing consistent business definitions, data types, and domain classifications across the analytics project.

Field definitions are sourced from the official **DescriptionDataCoSupplyChain.csv** reference file and cross-referenced with actual data values and standard supply chain terminology.

Scope

The dictionary documents **61 fields** across two datasets:

- **DataCoSupplyChainDataset.csv** — 53 fields covering customer, product, order, shipping, financial, and geospatial data.
- **tokenized_access_logs.csv** — 8 fields capturing e-commerce clickstream behavior data.

Document Structure

- **Section 3** — Business domain classification.
- **Sections 4–10** — Field dictionary grouped by domain.
- **Section 11** — Key relationships and foreign key mappings.
- **Section 12** — Reference files and metadata.

This document has a single responsibility: **define every field**. Data profiling, quality analysis, and business recommendations are covered in subsequent phases.

2. Dataset Overview

The following table summarizes the two source datasets documented in this dictionary.

Property	DataCoSupplyChainDataset.csv	tokenized_access_logs.csv
Rows	~180,519	~415,000
Columns	53	8
Primary Key	Order Id + Order Item Id (composite)	user_id + event_date (composite)
Foreign Keys	Customer Id, Product Card Id, Category Id, Department Id	order_id, item_id (link to main dataset)
Source System	ERP / Order Management System	E-commerce clickstream tracker

Reference File

File	Description
DescriptionDataCoSupplyChain.csv	Official field definitions. Two columns: FIELDS (exact column name) and DESCRIPTION (business definition). Primary reference for this dictionary.

3. Business Domain Classification

All 61 fields are classified into seven business domains. This grouping reflects how stakeholders reference data by business process.

Domain	Fields	Description
Customer	11	Customer identity, contact, and demographic information.
Product	11	Product catalog, category, and department hierarchy.
Order Transaction	14	Order placement, line items, quantities, discounts, and status.
Shipping & Logistics	8	Delivery timelines, shipping destinations, and tracking.
Financial Metrics	5	Revenue, profit, and margin calculations.
Geospatial	2	Latitude and longitude coordinates.
Clickstream	8	Web session tracking and user navigation behavior.
Total	61	

4. Customer Domain

Column Name	Business Definition	Data Type	Example
Customer Id	Unique identifier for each customer.	Integer	12345
Customer Fname	First name of the customer.	VARCHAR	John
Customer Lname	Last name of the customer.	VARCHAR	Smith
Customer Email	Email address associated with the customer account.	VARCHAR	john.s@email.com
Customer Password	Hashed credential for customer account authentication.	VARCHAR (hashed)	*****
Customer Segment	Customer classification (Consumer, Corporate, Home Office).	VARCHAR	Consumer
Customer Street	Street address for customer deliveries.	VARCHAR	123 Main St
Customer City	City of the customer.	VARCHAR	New York
Customer State	State or province of the customer.	VARCHAR	NY
Customer Country	Country of the customer.	VARCHAR	United States
Customer Zipcode	Postal code for the customer.	VARCHAR	10001

5. Product Domain

Column Name	Business Definition	Data Type	Example
Product Card Id	Unique identifier for each product (SKU-level).	Integer	456
Product Category Id	Identifier linking the product to its category.	Integer	12
Product Description	Free-text description of product features and specifications.	TEXT	Stainless steel bottle, 32oz
Product Image	URL for the product image displayed on the e-commerce site.	VARCHAR	https://.../img.jpg
Product Name	Display name of the product.	VARCHAR	Insulated Water Bottle 32oz
Product Price	Current list price of the product.	DECIMAL(10,2)	24.99
Product Status	Indicates whether the product is active, discontinued, or out of stock.	VARCHAR / INT	Active
Category Id	Unique identifier for the product category.	Integer	5
Category Name	Name of the product category.	VARCHAR	Camping & Hiking
Department Id	Unique identifier for the department.	Integer	2
Department Name	Name of the department.	VARCHAR	Outdoor Recreation

6. Order Transaction Domain

Column Name	Business Definition	Data Type	Example
Order Id	Unique identifier for each customer order.	Integer	56789
Order Customer Id	Foreign key to the customer who placed the order.	Integer	12345
order date (DateOrders)	Date and time the order was placed.	DateTime	2017-01-15 14:30:00
Order Item Id	Unique identifier for each line item within an order.	Integer	1
Order Item Cardprod Id	Foreign key to the product ordered.	Integer	456
Order Item Discount	Monetary discount applied to this line item.	DECIMAL(10,2)	3.50
Order Item Discount Rate	Discount rate applied to this line item (0.15 = 15%).	Float	0.15
Order Item Product Price	Unit price of the product at time of order.	DECIMAL(10,2)	24.99
Order Item Profit Ratio	Profit margin ratio for this line item.	Float	0.35
Order Item Quantity	Quantity ordered for this line item.	Integer	2
Type	Transaction classification (SALE, TRANSFER, PAYMENT).	VARCHAR	SALE
Order Status	Processing status of the order (PENDING, COMPLETE, CANCELED).	VARCHAR	COMPLETE
Sales	Total sales amount for this line item (Qty × Unit Price).	DECIMAL(10,2)	46.48
Order Item Total	Total amount for this line item after discounts.	DECIMAL(10,2)	46.48

7. Shipping & Logistics Domain

Column Name	Business Definition	Data Type	Example
Days for shipping (real)	Actual days from order placement to delivery.	Integer	5
Days for shipment (scheduled)	Planned delivery days at time of order placement.	Integer	4
Delivery Status	Current status of the delivery (Delivered, In Transit, Pending).	VARCHAR	Delivered
Late_delivery_risk	Binary flag: 1 if delivery was at risk of being late, 0 otherwise.	Integer (0/1)	1
Order City	Destination city for the shipment.	VARCHAR	Los Angeles
Order Country	Destination country for the shipment.	VARCHAR	United States
Order State	Destination state for the shipment.	VARCHAR	California
Order Region	Geographic region of the shipping destination.	VARCHAR	West

8. Financial Metrics Domain

Column Name	Business Definition	Data Type	Example
Benefit per order	Calculated profit contribution for the order.	DECIMAL(10,2)	15.75
Sales per customer	Total sales attributed to the customer across all orders.	DECIMAL(10,2)	450.00
Order Profit Per Order	Calculated profit for the order.	DECIMAL(10,2)	12.50
Order Item Profit Ratio	Profit margin ratio for line items.	Float	0.32
Market	Geographic market for the transaction (US, EU, LATAM, APAC).	VARCHAR	US

9. Geospatial Domain

Column Name	Business Definition	Data Type	Example
Latitude	Latitude coordinate of the shipping destination.	Float	40.7128
Longitude	Longitude coordinate of the shipping destination.	Float	-74.0060

10. Clickstream Domain

Column Name	Business Definition	Data Type	Example
event_date	Date and time the clickstream event was recorded.	DateTime	2017-06-15 10:23:45
click_stream	Encoded user browsing path during the session.	TEXT	home > search > product > cart
item_id	Identifier for the product viewed. Joins to Product Card Id.	Integer	456
user_id	Identifier for the web user.	Integer	12345
order_id	Order identifier if the session resulted in a purchase.	Integer	56789
product_name	Name of the product viewed during the session.	VARCHAR	Insulated Water Bottle 32oz
category_name	Category of the product viewed during the session.	VARCHAR	Camping & Hiking
access_token	Tokenized identifier for session tracking.	VARCHAR	a1b2c3d4e5

11. Key Relationships

The following table documents the primary foreign key relationships across the datasets.

Field	References	Relationship Type
Order Customer Id	Customer Id	Many-to-One (many orders per customer)
Order Item Cardprod Id	Product Card Id	Many-to-One (many line items per product)
Product Category Id	Category Id	Many-to-One (many products per category)
Category Id → Department Id	Department Id	Many-to-One (many categories per department)
item_id (clickstream)	Product Card Id (main dataset)	Many-to-One (many views per product)
order_id (clickstream)	Order Id (main dataset)	Many-to-One (many events per order)
user_id (clickstream)	Customer Id (main dataset)	Many-to-One (if authenticated)

Composite Key Note

In DataCoSupplyChainDataset.csv, **Order Id + Order Item Id** form a composite key uniquely identifying each row (one row per line item). This is critical for ensuring accurate aggregations and avoiding double-counting in revenue and profitability analysis.

12. References

The following files and resources were used to compile this data dictionary.

Reference	Description
DescriptionDataCoSupplyChain.csv	Official field definitions from the dataset publisher. Contains FIELDS (column name) and DESCRIPTION (business definition) for all 53 columns in the main dataset.
DataCoSupplyChainDataset.csv	Primary dataset: 53 columns, ~180,519 rows. Order, customer, product, shipping, and financial transaction records.
tokenized_access_logs.csv	Supplementary dataset: 8 columns, ~415,000 rows. E-commerce clickstream tracking data.

End of Enterprise Data Dictionary. Proceed to Phase 4: Data Profiling & Quality Analysis.